

Python Assignment Questions

Assignment 1: Python Basics

1. Print 'Hello, World!'.
2. Perform arithmetic operations.
3. Swap two variables.
4. Find whether a number is even or odd.
5. Find the largest of three numbers.

Assignment 2: Control Statements

1. Check whether a year is a leap year.
2. Generate the multiplication table of a number.
3. Find the factorial of a number.
4. Generate the Fibonacci series.
5. Check whether a number is prime.

Assignment 3: Functions

1. Create a function to calculate the square of a number.
2. Write a recursive factorial function.
3. Write a recursive Fibonacci function.
4. Demonstrate lambda functions.
5. Use map(), filter(), and reduce().

Assignment 4: Strings

1. Reverse a string.
2. Check for palindrome.
3. Count vowels and consonants.
4. Check whether two strings are anagrams.
5. Count character frequency.

Assignment 5: Lists, Tuples, Sets & Dictionaries

1. Find the largest element in a list.
2. Remove duplicates using a set.
3. Sort a list.
4. Create and manipulate dictionaries.
5. Merge two dictionaries.

Assignment 6: File Handling

1. Write to a text file.

2. Read from a text file.
3. Copy one file to another.
4. Count words and lines in a file.
5. Append data to an existing file.

Assignment 7: Exception Handling

1. Handle division by zero.
2. Use try-except-finally.
3. Raise a custom exception.
4. Handle multiple exceptions.
5. Demonstrate assert.

Assignment 8: OOP

1. Create a Student class.
2. Demonstrate constructors.
3. Implement inheritance.
4. Demonstrate polymorphism.
5. Demonstrate encapsulation.

Assignment 9: Modules & Packages

1. Create a custom module.
2. Import and use built-in modules.
3. Use the math module.
4. Use the random module.
5. Create a package.

Mini Projects

1. Student Management System
2. Library Management System
3. Bank Management System
4. Quiz Application
5. To-Do List App
6. Contact Book
7. Expense Tracker
8. Inventory Management System
9. Password Generator
10. Weather App using API